

New OLED TDDI Driver IC Architecture

TsangChih WU, Huajun Zhuang

Display & Touch Department, Xiaomi Co.Ltd. Beijing, China

Abstract

This paper presents an integrated OLED TDDI driver IC architecture featuring mutual-capacitance touch and proprietary MUX-in-IC design. Addressing OLED's core challenges of cost, form factor, and power consumption, it reduces chip area and system power significantly while enabling ultra-narrow bezels, paving a viable path for TDDI adoption in OLED displays.

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1. Introduction

Since Synaptics introduced the first LCD TDDI in 2014, LCD display technology has undergone a rapid transition over the past decade, moving from discrete DDIC and TP IC solutions to an integrated TDDI solution. Leveraging significant advantages in cost and power consumption, LCD TDDI quickly reshaped the market. It now holds a dominant position in the mobile phone sector and has begun expanding into automotive display applications since 2024.

System integration is a key path for technological evolution. However, in the OLED field, display and touch have long adhered to a separated architecture involving a discrete display driver IC and a touch controller IC. The industry has consistently failed to achieve mass production of a TDDI solution, fundamentally because the core value proposition for OLED TDDI remained unproven.

This article systematically introduces Xiaomi's core development strategy in display driver chips, focusing on the following areas:

1. The technology roadmap and overall strategy for display driver ICs led by Xiaomi
2. The technical solution and plan for the Single RAM DDIC
3. The chip architecture, technical evaluation, and development of OLED TDDI
4. The development blueprint for next-generation OLED display driver ICs

2. Single RAM DDIC: Foundational Innovation

Before detailing the OLED TDDI solution, it is essential to clarify its critical technological foundation—the Single RAM OLED DDIC developed by Xiaomi. This innovation began with design evaluation in 2021 and successfully achieved technology announcement and mass production in 2022. Its core value lies in fundamentally restructuring the traditional architecture to resolve the conflict between high performance and high cost.

The conventional Dual RAM OLED DDIC architecture incorporates two independent SRAMs in its digital section: one for frame data buffering and another for screen uniformity

compensation (Demura). Its analog circuit uses a MUX 1:1 source driver design, where each source data line corresponds to a dedicated driver amplifier. While ensuring performance, this architecture leads to a significant increase in chip area and power consumption.

Xiaomi's Single RAM DDIC solution achieves a breakthrough through a series of co-design innovations:

Removal of the Frame Buffer SRAM: Enables real-time data processing without a dedicated frame buffer

Restructured Source Driving: Upgrades the driver circuit to a MUX 1:2 design, allowing one source driver amplifier to service two data lines, effectively halving their count

Collaborative Panel Design: Promotes panel adoption of a Dual Data Line (DDL) architecture to match the new driving method, ensuring data bandwidth for high refresh rates

This architectural transformation delivers direct and substantial benefits: the chip area is reduced by over 46% compared to the traditional Dual RAM solution, significantly lowering wafer costs. This provides crucial support for cost-effective mass production of OLED display driver chips and lays a solid foundation in area and cost for subsequent touch integration.

3. OLED TDDI Integration Challenges

The integration of display and touch for OLED represents a major technological challenge following the success of LCD TDDI. Its development logic shares similarities with the LCD path but involves fundamental differences.

In the LCD domain, evolution is clear: it transitioned from separate Display Driver ICs and Touch ICs (requiring an external TP sensor, a discrete TP IC, and a TP FPC) to a single-chip TDDI solution integrating touch into the display driver. This shift not only reduced IC cost but also, by eliminating the external touch sensor, delivered significant system-level advantages like module-level cost reduction, thinner modules, and supply chain simplification. This gained strong momentum from panel makers and brands, making it mainstream rapidly.

However, directly applying this model to OLED faces inherent obstacles. Mainstream OLED panels already use a TP on Cell structure (touch sensor fabricated on the encapsulation glass). Therefore, adopting OLED TDDI cannot achieve the same "module-level" cost saving by removing an external sensor as LCD TDDI did. Consequently, the initial value proposition for OLED TDDI needed redefinition, and its development lacked clear module-level impetus, progressing slowly.

Based on industry insights, Xiaomi began pre-research on OLED TDDI in 2021. Initial feasibility assessment was based on the NT78770 reference design (self-capacitance touch, dual-RAM architecture), completing validations including module process, reliability, and COB packaging. Technical evaluation revealed this architecture, due to integrating dual RAMs (frame buffer and Demura SRAM) and using a conventional design,

resulted in a chip height of 1970 μ m and an area about 2.3 times that of a standard single-RAM DDIC. From a cost-benefit perspective, the large chip size led to prohibitively high wafer costs, making it unsuitable for mass production.

Facing this bottleneck, Xiaomi's R&D team leveraged in-depth pre-research data to innovatively restructure the system architecture, proposing the Single RAM TDDI (Mux in IC) integrated solution. Its core breakthroughs are:

Architecture Simplification: Removing the dedicated frame buffer SRAM and optimizing data flow

Circuit Innovation: Changing the source driver design from traditional MUX 1:1 to MUX 1:2

Deep Integration: Integrating the MUX switch circuit, originally on the panel, into the driver IC

This solution achieves significant reduction in chip area and power consumption while maintaining touch performance and not adding complexity to existing module manufacturing processes, providing a viable path for cost-effective mass production of OLED TDDI.

4. Touch Solution: Mutual Capacitance Optimization

Touch Solution Selection: From Path Dependency to Precise Adaptation. A primary technical decision in defining the TDDI solution is choosing between self-capacitance or mutual-capacitance touch sensing. This choice directly impacts end-product touch performance, form factor, and manufacturability.

LCD TDDI widely adopted self-capacitance for its high report rate and compatibility with panel ITO design. Thus, some manufacturers initially followed this path in early OLED TDDI exploration.

However, Xiaomi's validation of self-capacitance OLED TDDI since 2021 revealed significant bottlenecks for OLED:

1. Limits Narrow Bezel Design: Self-capacitance requires independent routing per touch electrode, increasing trace density and width at panel edges, hindering narrow bezels contrary to OLED's goal of extreme screen-to-body ratio

2. Increases Test Complexity/Cost: It needs more touch test pads, encroaching on limited space near the chip. For designs placing test circuits under the DDIC, this adds layout challenges and cost. Shrinking pads conflicts with existing production line test equipment, raising barriers

Based on this analysis, we determined mutual-capacitance is superior for OLED TDDI. It inherently offers high SNR, strong anti-interference, and easier narrow bezel implementation, aligning better with OLED's technical traits and premium positioning. Concurrently, we migrated touch digital circuits from 55nm to more advanced 40nm or 28nm process nodes. This upgrade alone brings about 10-15mW of static power saving for the touch subsystem, optimizing both performance and efficiency.

		mutual-cap	Self-cap
TP performance	RC	350pF	50pF
	SNR	40dB	40dB
	report rate	240Hz@10f 360Hz@2f	360Hz@10f
	LGM	☆☆	☆☆☆
	Waterproofing	☆☆	☆☆☆
	Active pen	support	support

5. Mux-in-IC Architecture

The Mux in IC solution is a Xiaomi proprietary patent obtained in 2020. Its core innovation is redesigning and integrating the analog switch (MUX) circuit, traditionally on the display panel, into the display driver chip. This architectural change delivers significant multi-dimensional benefits at the system level for power, space, and EMI.

Significant Power and Efficiency Optimization

Circuit Simplification: Integrating MUX from panel to DDIC halves the number of source operational amplifiers versus traditional "MUX 1:1" architecture, greatly reducing analog circuit static power.

Improved Drive Efficiency: Internal switches have better resistance; their equivalent RC load is much smaller than panel traces. Thus, operating voltage drops drastically from panel-side high/low voltages (VGH/VGL, typically ± 7 -15V) to a mid-voltage of 8V. This reduces conversion loss and boosts overall drive efficiency.

Achieving Ultra-Narrow Bezel

Frees Panel Space: Traditionally, on-panel MUX circuit height is ~ 250 -330 μ m, a key border width constraint. Integrating MUX into IC frees this routing area, narrowing display side borders by ~ 200 μ m, directly enabling higher screen-to-body ratio.

Enhanced Electromagnetic Compatibility

Built-in Shielding: The high-speed switching MUX is a potential EMI source. Integrating it into the chip enables near-field shielding via the IC's silicon substrate and metal layers. Combined with external EMI tape, it forms "on-chip + module-level" dual shielding, more effectively suppressing MUX-induced radiation noise and improving overall device EMI performance.

In summary, Mux in IC is not mere function relocation. It's a key technology achieving comprehensive optimization of power, form, and reliability at the system level through chip-panel co-design.

Process Selection

For the first mass-produced OLED TDDI IC targeting 1.5K resolution mid-range modules—a highly cost-sensitive segment—extreme cost optimization was a core design strategy. Process node choice was key.

During evaluation, a comprehensive comparison was made between mainstream 55nm and more advanced 40nm processes, assessing capacity and wafer cost, and process traits and chip area.

Detailed modeling showed 40nm offered superior overall cost-effectiveness. While its per-wafer cost might be slightly higher than mature 55nm, its higher transistor density and better efficiency allowed implementing all functions on a smaller chip. The significant area reduction directly and substantially offset the wafer cost difference.

Final calculations indicate the 40nm OLED TDDI single-chip solution achieves ~10% lower overall system cost versus the traditional "discrete DDIC + discrete TP IC" dual-chip solution. This met strict cost control requirements while ensuring sufficient market competitiveness in performance and power efficiency.

6. Process Selection for Cost Optimization

Targeting cost-sensitive mid-range 1.5K resolution modules, a 55nm vs. 40nm process comparison prioritized wafer cost and supply capacity, and transistor density, performance, and area impact.

Modeling confirmed 40nm's superior cost-effectiveness: higher density offset slightly higher wafer costs, enabling a ~10% lower system cost than discrete dual-chip solutions while meeting performance and power requirements.

7. Summary

Over five years, we completed the full-cycle endeavor from technical pre-research to large-scale mass production of the OLED TDDI solution. The breakthrough core is the pioneering "Mutual Capacitance + MUX in IC" integrated architecture.

Through profound technological integration and architectural innovation, this solution has delivered significant multidimensional benefits: it saves approximately 30 mm² of valuable space in the overall layout, creating more possibilities for the battery or rear chamber module; aids in reducing the side border width by 0.1–0.15 mm, directly enhancing the screen-to-body ratio and visual immersion; improves energy efficiency by lowering power consumption by 45 mW, effectively extending device battery life; and streamlines system design by eliminating the need for both the TP IOVCC and TP VCI power supplies, thereby simplifying peripheral circuitry and reducing both Bill of Materials (BOM) cost and design complexity. Furthermore, the solution optimizes signal integrity, enhancing anti-interference capability.

By innovating at the architectural level, this OLED TDDI solution breaks the traditional zero-sum game between IC performance, power consumption, and size. It successfully achieves the dual objectives of cost reduction and power optimization, positioning itself to pioneer a new direction for innovation within the industry.

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9. References

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